Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

1. Background/Introduction

1.1 Relevance to DOE Genesis Mission

We are responding to "The Genesis Mission: Transforming Science and Energy with AI." In particular we address project topic 21, "Artificial Intelligence in Fluid Flow for Energy Components and Technologies." The focus area is 21A, "Physics-Informed AI for complex fluid phenomena."

This proposal requests funding in order to develop improved numerical methods, via AI enhancement, in order to simulate atomization and spray processes as they pertain to combustion in aircraft engines[4] and rotating detonation engines[6, 3] used by rockets.

As a group, the PIs Sussman, Shoele, and Arienti have developed over time state of the art adaptive, parallel, exascale enabled, numerical algorithms for carrying out high-fidelity numerical simulations of atomization and spray and related problems[4, 5, 3, 6, 29, 17, 30, 27, 21].

This all being said, there are limitations in **exascale CFD**, where multiscale interfacial physics imposes severe computational constraints[14]. It is proposed herein to develop an efficient computational system in which existing high-fidelity simulation and experimental data are used, sparingly, in order to develop sub-scale models which can then be used to solve a more general set of atomization and spray problems (e.g. varying the geometry, fluid physical properties, and boundary conditions).

Consistent with Topic Area 21, "Physics-Informed AI for complex fluid phenomena," we shall (1) develop the capability to assimilate data from high fidelity simulations and/or experiments into our existing foundational methodology for multiphase flow[4, 5, 3, 6, 29, 17, 30, 27, 21], and (2) use AI in order to learn sub-scale models for predicting the difference between underresolved simulations and super-resolved high fidelity simulations (and/or experiments).

The goal is to develop an AI-generated sub-scale model in order to minimize the frequency of having to run high-fidelity simulations.

We remark that the benefits of this proposal can be broad. In energy applications, accelerated multiphase simulation enables **improved predictive capability** for fuel injection, combustion processes, systems with boiling, evaporation, and condensation, and advanced propulsion systems. The integration of simulation with data-driven methodologies supports the development of **adaptive digital environments** in which modeling, data, and computation are tightly coupled to *accelerate discovery and innovation*.

1.2 Proposed Research in the context of the current state-of-the-art

Multiphase flows with *sharp, dynamically evolving interfaces* play a central role in energy conversion and transport systems, including fuel injection, spray combustion, rotating detonation engines, and phase-change-driven thermal processes[3, 5, 4, 6, 14, 29]. These flows are governed by *tightly coupled nonlinear interactions* among turbulence, interfacial forces, thermodynamics, and material response across a wide range of spatial and temporal scales, and are critical for identifying distinct mass and momentum processes at evolving interfaces. Their accurate prediction remains **one of the most challenging problems in computational science**.

Over the past several decades, semi-implicit **sharp-interface methods** have emerged and matured as a rigorous framework for modeling such systems with provable performance. Foundational work by [26] established level set methods for capturing evolving interfaces in multiphase flow. Subsequent developments led to conservative formulations[25] that significantly improved *mass conservation and robustness* in the presence of topology change.

A major advance in this area is the **moment-of-fluid (MOF)[9, 1, 12]** and cell-integrated semi-Lagrangian **compressible MOF framework developed by Jemison, Sussman, and Arienti[11]**. The compressible multiphase formulation developed by Jemison et al introduced a unified method that conserves mass, momentum, and energy while capturing shocks and material discontinuities without Riemann solvers. The material interfaces in [11] are represented sharply while at the same time preserving volume to machine precision in zero mach number regions. Our approach enables simulations across compressible and incompressible fluids through asymptotic-preserving semi-implicit pressure updates. Further developments demonstrated accurate treatment of *multi-material junctions* and surface tension effects in complex geometries[6, 3, 17, 27, 13, 30, 29].

In parallel, high-fidelity studies of fuel injection and atomization using DNS and LES approaches have highlighted the *extreme computational demands* of resolving interface breakup, turbulence interaction, and compressibility effects in realistic engine conditions and, more importantly, for problems with sharp mass transfer at the interface. In the article by Li and Soteriou[14], the authors demonstrate that, for simulating the atomization of a liquid jet in a cross flow (JICF), adaptive mesh refinement provides only a marginal benefit for practical multiphase flow problems. The JICF problem exhibits a feedback mechanism between the chaotic gas flow and the breakup of the liquid jet. This motivates new approaches to improve the solution procedure employed in these methods while not abandoning their unique predictive capabilities.

One possible avenue of improvement that we had considered, as well as that considered by others[30, 16], was to replace costly geometric operations in multiphase flow algorithms with surrogate AI-generated models. The key principle is that surrogate models must not compromise on the fidelity of the numerical algorithm. In particular, for volume tracking, or moment of fluid methods, it is essential that the "linearity preserving" and "volume preserving" properties of a multiphase flow system not be violated[9, 1, 23, 22, 24]. Yet, what was discovered by us[30], and also mentioned in[16] is that these AI generated surrogate algorithms, at best, provide a marginal benefit. We also bring attention to the following articles which give evidence that surrogate AI models will never beat finely tuned optimized algorithms[18, 7, 2, 15].

Another possible avenue of improvement is to develop a meshless method such as PINN[19, 10] or SPH[28, 20]. These meshless methods, or any other pure meshless method, are not volume preserving. In otherwords, in the limit as the mach number approaches zero, the volume should be preserved. We note that the volume error, using PINN, reported by [19] was *worse* than even that of classical level set methods.

It is proposed herein to make two new developments: (I) we shall improve our existing compressible multiphase flow algorithm[11, 3, 6] by developing an accompanying volume and mass preserving compressible multiphase flow nudging capability for the assimilation of multiphase flow data from either high-resolution, high-fidelity simulations, and/or experiments[8]. (II) We shall use the nudging correction generated from (I) to train a neural network to derive sub-scale models appropriate for underresolved multiphase flows. It is expected that our sub-scale model will be general for a wide

range of multiphase flows.

2. Project Objectives

The project objectives address project topic 21, "Artificial Intelligence in Fluid Flow for Energy Components and Technologies." The focus area is 21A, "Physics-Informed AI for complex fluid phenomena."

The project objectives are to (I) improve our existing compressible multiphase flow algorithm[11, 3, 6] by developing an accompanying compressible multiphase flow nudging capability for the assimilation of multiphase flow data from either high resolution, high fidelity simulations, and/or experiments. The nudging algorithm we propose to develop will enable volume and mass preserving "interface nudging" suitable for the complex atomization and spray problems that we consider. We propose to implement the algorithm developed Dommermuth et al[8]. In (31) of [8], the nudging term is $-\frac{p_s}{\rho}\nabla H$ in which p_s is a nudging stress that acts normal to the interface. p_s is proportional to the difference between the simulation level set function and an assimilated level set function. (II) We shall use the nudging correction, generated from (I), in order to train a neural network in order to derive sub-scale models appropriate for underresolved multiphase flows. It is expected that our sub-scale model will be general for a wide range of multiphase flows.

Our targeted canonical problems are (a) the atomization of a fuel Jet in a Cross Flow problem (JICF) which address combustion in aircraft engines[4] and (b) the shock cylinder/drop problem which address rotating detonation engines used by rockets[6, 3].

3. Proposed Research and Methods; Physics-Constrained Data Driven Model

The proposed research builds directly on prior work by PIs **Sussman, Shoele and Arienti[11, 27, 17, 30, 29]**, particularly the *cell-integrated semi-Lagrangian framework* and *moment-of-fluid interface representation*, which enable accurate tracking of deforming interfaces and efficient handling of *compressibility and stiffness*. Materials are represented using Volume fractions (zero order moment) and centroids (first order moment). Interfaces are captured. The governing equations include conservation of mass, momentum, energy, and species, with extensions for viscoelasticity, elasto-plasticity, and phase change. Interface conditions enforce *stress balance, velocity discontinuities due to mass transfer, and thermodynamic equilibrium*. Surface tension effects are modeled through *curvature-dependent forces*, and phase change is driven by *temperature and concentration gradients and corresponding nonlinear surface mass and chemical transport equations*.

In the proposed physics-constrained, solver-embedded learning framework, (1) high fidelity computational data and/or experimental data are assimilated via nudging into a coarse-grid simulation of two-phase flows, and (2) unresolved turbulence and interfacial dynamics are modeled using neural networks trained on the nudging source terms. Unlike conventional Physics-Informed Neural Network approaches that attempt to directly approximate the solution field, the present method retains the full structure of a conservative MOF-based two-phase flow solver, ensuring sharp interface representation, strict mass conservation, and compatibility with established numerical methods. The neural network is instead tasked with learning closure interfacial and bulk force corrections which will be embedded directly into the governing equations at each time step. Crucially, additional interface-aware constraints ensure that corrections remain localized and consistent with geometric

quantities such as interface normals and curvature, addressing a major limitation of existing data-driven models in multiphase flows. When high-fidelity data are available, a compressible multiphase data assimilation nudging technique is used to align the coarse solution with reference dynamics, enabling accurate reconstruction of key flow features at a fraction of the computational cost. We note that when material interfaces are nudged, we ensure volume and mass preservation using the algorithms developed by Dommermuth et al[8] (see their equation (31)).

In this work, since independently evolved coarse-grid and high-fidelity two-phase simulations can rapidly diverge due to nonlinear amplification of interfacial errors, the raw mismatch between them does not provide a clean learning target for unresolved physics. Instead, the coarse solution is periodically assimilated toward the restricted high-fidelity state, $Q^{n+1}_{hf\to c}$, and the resulting aligned coarse state is used as the initial condition for one-step coarse-grid advancement. The discrepancy between this uncorrected coarse update and the corresponding high-fidelity evolution defines the target correction for the neural network. This strategy ensures that the learned model captures missing subgrid and interfacial physics, rather than compensating for accumulated trajectory drift.

The neural network is trained to infer unresolved interfacial and near-interface corrections from coarse-grid flow fields and reconstructed interface features, enabling the coarse solver to recover high-fidelity dynamics that are otherwise lost at reduced resolution.

Let Q^n_a denote the assimilated coarse state at time t_n, and let $Q^{n+1}_{c,unc} = G_{\mathrm{MOF}}(Q^n_a)$ be the corresponding uncorrected one-step coarse update. The target correction is defined as the one-step discrepancy between the restricted high-fidelity evolution and the uncorrected coarse prediction, $F^n_{\mathrm{corr}} = (Q^{n+1}_{hf\to c} - Q^{n+1}_{c,unc})/\Delta t$. A neural network $\mathcal{N}_\theta$ is then trained to approximate the mapping from coarse-grid flow and interface features $z^n = \Phi(Q^n_a, \Gamma^n_a)$ to this correction, where Γ^n_a is obtained from the MOF reconstruction of the assimilated moments. The training objective minimizes the discrepancy between predicted and target corrections, $\mathcal{L}(\theta) = \sum_n \|W \cdot (\mathcal{N}_\theta(z^n) - F^n_{\mathrm{corr}})\|^2 + \mathcal{R}(\theta)$, incorporating optional weighting W and regularization $\mathcal{R}(\theta)$ concentrated near the interface.

Because the dominant modeling errors in coarse-grid two-phase simulations occur in mixed cells and in a narrow band surrounding the reconstructed interface, both supervision and regularization of the neural network are concentrated in these regions during training. Specifically, (1) the supervised loss is weighted by an interface-aware indicator so that discrepancies in interfacial regions contribute more strongly than errors in the bulk phases, where the coarse solver is typically already accurate. This weighting can be implemented using either a binary interface mask or a smooth distance-based function that increases the contribution of cells closer to the interface. (2) Physically motivated regularization terms are introduced to constrain the structure of the learned correction: 2a: A localization penalty is applied to discourage the network from producing spurious corrections away from the interface. 2b: A smoothness regularization is imposed within the interface band to prevent nonphysical oscillations and enforce spatial correction along the interface. 2b: A magnitude constraint is included to bound the correction and avoid overfitting or instability.

For interfacial force corrections, the third regularization term (3) is an additional geometric regularization that enforces alignment with the reconstructed interface normal, ensuring consistency with the underlying physics of interfacial interactions.

Together, these interface-focused weighting and regularization strategies ensure that the neural network learns physically meaningful, localized corrections associated with unresolved interfacial and subgrid-scale processes, while avoiding contamination from bulk-region noise or unphysical

behavior.

4. Milestones in the Nine Months

1. (Sussman Months 1-3): Enhance the existing compressible MOF algorithm[11, 3, 6, 29] by adding the ability to assimilate, via nudging, compressible multiphase interface, velocity, and temperature data. Volume and mass conservation is ensured during the nudging procedure via the algorithm reported in earlier work by Dommermuthg et al[8].

2. (Arienti Months 1-3): Develop routines that import high-resolution data[3, 6] into actively running coarse grid simulations. Write restriction routines which carry out the operation: $Q^{n+1}_{hf\to c}$.

3. (Shoele Months 1-3): Develop the neural network algorithm which minimizes the discrepancy between predicted and target corrections, $\mathcal{L}(\theta) = \sum_n \|W \cdot (\mathcal{N}_\theta(z^n) - F^n_{\text{corr}})\|^2 + \mathcal{R}(\theta)$, incorporating optional weighting W and regularization $\mathcal{R}(\theta)$ concentrated near material interfaces.

4. (Arienti, Sussman, Shoele Months 4-6): Integrate the developments in items 1-3 above: nudging data assimilation, AMReX data base management, AI generated sub-scale models. Diagnostic testing on 2D atomization and spray from shock-cylinder interaction. Verification and validation.

5. (Arienti, Sussman, Shoele Months 7-9): Same as task 4, except fully 3D problems will be investigated)

5. Data Sources and Models

The current version of the computer code that is used to generate results shown in Figure 3 of [4], Figures 4 and 6 of [3] and Figures 5 and 11a of [6] is stored on Git Hub:
`https://github.com/msussman42/amrex_implicit_interfaces.git`
The data in Figure 3 of [4], and Figure 4 of [3] can be regenerated from scratch within 30 days. The remainder of the data is stored by Marco Arienti on national Lab computers.

6. Decision Gate Metrics

We shall train our neural network subscale models on the 2D/3D water and Galinstan shock-cylinder interaction problems (Figures 4,6,7 of [3]). The effective fine grid resolution for the "high fidelity" simulation is $3072 \times 2304 \times 768$ (3D). The effective fine grid resolution for the "low fidelity" simulation is $384 \times 288 \times 96$. After training, we shall rerun the water and galinstan cases, except using the trained models instead of assimilating the "high fidelity" data. Then we shall compare the respective interfaces between (a) coarse grid, assimilated results, (b) coarse grid, model results, and (c) high fidelity results. Finally, we shall create two new cases, varying the Weber number and Reynolds number. For each of the new cases we shall compare to the high fidelity model: (a) coarse grid, no model, (b) coarse grid with model. The interface errors are computed using the symmetric difference error[12].

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project summary
phase I
Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

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The atomization and evaporation of fuel sprays is a fundamental process that takes place in internal combustion engines. This proposal focuses on new numerical methods for carrying out "digital twin" atomization experiments, leading to the improvement in the efficiency and/or performance of jet engines and rotating detonation rocket engines.

Effective Numerical methods and experiments alike must reliably measure multiscale processes ranging from millimeters to nanometers. Macro scale sprays break up into sheets and ligaments, which in turn break up into a mist. Evaporation turns the mist into a gaseous fuel which is ready to be reacted with oxidizer. The spray breakup process, along with convective heat transfer from evaporation and preceding reactions, induces a feedback mechanism in which the resulting turbulent flow field affects the breakup. At the same time, for rotating detonation engines, not all of the droplets need to be atomized prior to shock wave interaction. Remaining micro-scale sized droplets that survived the "pre-shock" atomization process are shattered by the supersonic detonation wave into a sub-micron "fog" almost instantly. At which point reactions occur, thereby generating the detonation wave for the next cycle. It is noted that in an optimized rotating detonation engine, the "pre-shock" remaining droplets must not be too large (so as to survive the shock wave interaction) and not too small either (so as not to have used too much energy in the pre-shock atomization phase).

It is with this backdrop that the reality is that both high-fidelity atomization and spray simulations and physical experiments are very expensive. This proposal's objective is to develop novel Artificial Intelligence (AI) methods for making the best use of data that is available. The proposed work consists of 3 parts: (1) data assimilation, (2) AI learning of multiphase flow sub-scale models, and (3) verification and validation.

1. Recent multi-phase flow data assimilation algorithms have been developed by Dommermuth et. al. for assimilating measured sea surface data for the purpose of improving incompressible multiphase flow predictions as well as enabling the discovery of hidden experimental states. We shall develop analogous data assimilation techniques with the extended capability for compressible multiphase flows.

2. The discrepancy between a coarse grid simulation, with data assimilation, and the corresponding high fidelity data, is used as training data for a neural network model in order to learn the coarse grid correction term, using solely coarse grid simulation data.

3. Coarse grid simulations are carried out for atomization and spray cases not included in the original data assimilation process (step 2). The coarse grid simulations are carried out with/without the aid of the neural network model developed in case 2 above.